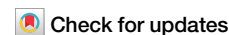


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Solvent- and metal-free upcycling of low-density polyethylene using a practical ZSM-5/Al₂O₃ bead catalyst



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Upcycling plastic waste into valuable resources without solvents, precious metals, or additional H₂ under mild conditions is a formidable challenge that requires new innovations. Herein, we document a catalyst based on Al₂O₃ beads and zeolite ZSM-5 (ZSM-5#Al₂O₃(2 h)). It cracked polyethylene (PE) into gasoline-range C₄ – C₁₂ hydrocarbons with a selectivity of 98% and a yield of >71% within 1.5 h at 260 °C. Characterization of the catalysts revealed a mesoporous structure and suitable Brønsted acid sites, which promoted the diffusion and catalytic activation of PE. Adjusting the dosage of zeolite supported on Al₂O₃ beads enables regulation of both the number and strength of acidic sites. Theoretical calculations confirmed that the interfacial interaction between zeolite ZSM-5 and Al₂O₃ reduces the acid strength of the composite. This is beneficial for the formation of higher hydrocarbons. Remarkably, the process was performed in a 1 L CSTR unit. These results provide instructive guidance for the design of high-performance catalytic materials used in PE waste recycling.

Since their invention in the 19th century¹, plastics have transformed modern life; however, plastic waste has become a critical environmental problem². Current plastic waste handling methods are largely unsustainable, with 50% of waste being landfilled, 19% incinerated, 22% discarded, and 9% recycled worldwide³. Mechanical recycling is attractive but requires specific raw materials and yields low-value products⁴, whereas chemical recycling serves as an attractive complementary approach. It can facilitate the selective recycling of plastic waste into high-value fuels and basic chemicals^{5–7}.

Catalytic pyrolysis, a chemical recycling method, allows for the use of lower temperatures and is more suitable for many industrial processes⁸. The type of catalyst is key to the efficient conversion of plastic waste into valuable products. Solid acid catalysts (including zeolites, metal oxides, and SiO₂-Al₂O₃) are widely used, as expected from their proven performance in crude oil refining^{9–12}. The cracking of PE over solid acid catalysts is influenced by their acidity and pore size. The ZSM-5 zeolite, with its ordered pores, strong Brønsted acidity, and thermal stability, is a suitable catalyst for the catalytic pyrolysis of PE^{13,14}. However, its microporous structure restricts the diffusion of pyrolysis intermediates throughout the pores (micropores are < 2 nm), resulting in an increase in the gas product (C₁–C₃) yield¹⁵.

To increase the selectivity for liquid products (e.g., C₄–C₁₂), mesopores (2–50 nm) can be introduced into catalysts to improve the diffusivity of components¹⁶. This can be performed using top-down and bottom-up

methods^{6,17,18}. These methods enhance access to active acidic sites and facilitate the formation of low-molecular-weight substances. However, they partially destroyed the zeolite structure and involved expensive templating agents^{19,20}. The introduction of zeolites into mesoporous materials (based on SiO₂ and Al₂O₃) has also been explored to introduce mesopores and tune the distribution of acid sites²¹. In one of our related studies, a zeolite-modified Al₂O₃-bead catalyst (ZSM-5/Al₂O₃-b) was prepared²². This catalyst exhibited much higher selectivity for pre-cracking products and superior zeolite dispersion compared to the mechanically mixed catalysts (ZSM-5 (5%) + Al₂O₃). Recently, a strategy based on the low-temperature melting of PE and catalysis has gained significant attention^{2,23}. With this approach, PE can be pyrolyzed into valuable light aromatics. H₂ was generated in situ on the zeolite, and molten PE promoted effective contact with the catalyst². A gap in the literature is that only a few studies have investigated polyolefin pyrolysis using composite catalysts under mild conditions (<300 °C). Thermodynamically, the positive entropy change (ΔS > 0) during depolymerization lowers the overall Gibbs free energy of the system (ΔG = ΔH – TΔS)²⁴. Consequently, this trend promotes the extensive dispersion of low-molecular-weight species at elevated temperatures, effectively shifting the equilibrium toward fragmented products. Achieving the desired selectivity requires low-temperature depolymerization under kinetic control, coupled with the rational design of catalytic materials and reaction

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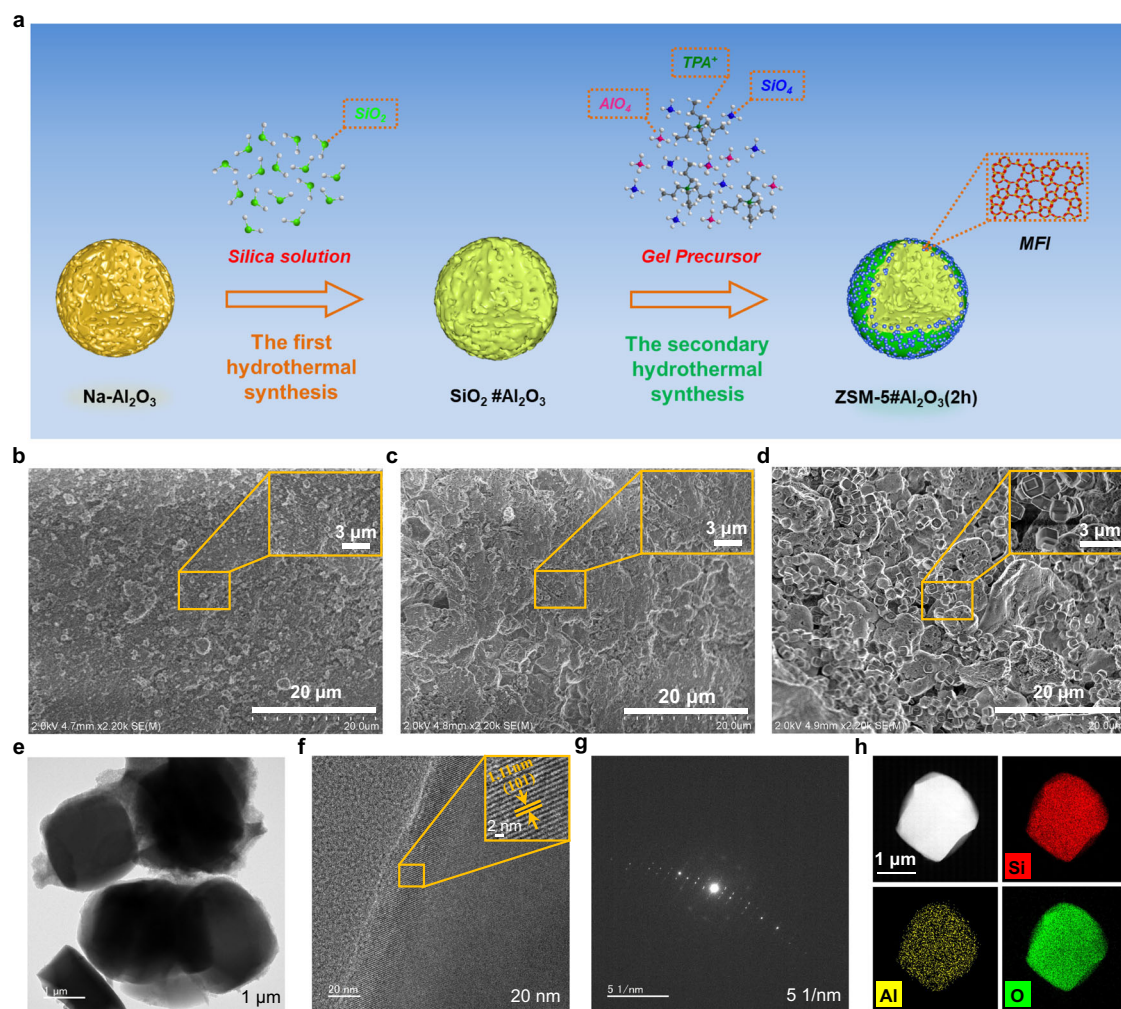


Fig. 1 | Morphology of the catalyst. **a** Schematic illustration of the ZSM-5/ $\text{Al}_2\text{O}_3(2\text{ h})$ sample preparation. **b–d** SEM images of the as-prepared samples. **b** Al_2O_3 . **c**, $\text{SiO}_2/\text{Al}_2\text{O}_3$. **d**, ZSM-5/ $\text{Al}_2\text{O}_3(2\text{ h})$. Macropores on the surface of ZSM-5/ $\text{Al}_2\text{O}_3(2\text{ h})$ resulted from alkali etching and secondary crystallization.

e–h Electron microscopy images of the powders scraped from the ZSM-5/ $\text{Al}_2\text{O}_3(2\text{ h})$ surface. **e** A low-resolution TEM image. **f** A high-resolution TEM image. **g** Selected-area electron diffraction (SAED). **h** TEM-EDS elemental mapping images.

systems^{24,25}. Adopting mild reaction parameters facilitates the precise modulation of product distribution by suppressing non-selective thermal degradation and favoring specific catalytic pathways. Low temperature is important, and catalytic design is critical.

To address this, a supported catalyst was designed and developed by growing ZSM-5 zeolites on Al_2O_3 beads via a two-stage hydrothermal synthesis. The preparation method and synthesis mechanism of the catalyst are comprehensively described and discussed. The catalytic activity of the supported catalyst for upgrading low-density PE (LDPE) to a gasoline-like fuel was systematically evaluated, and a comparative analysis was conducted with the physically mixed catalyst.

Results and Discussion

Synthesis and characterization

The approach used to prepare ZSM-5/ $\text{Al}_2\text{O}_3(2\text{ h})$ is shown in Fig. 1a. Spherical particles of Al_2O_3 (diameter of approximately 2 mm) with amorphous nanoclusters (Fig. 1b and Supplementary Fig. 1a) were used as the support phase. In contrast to the porous support (Al_2O_3 beads), the surface of the $\text{SiO}_2/\text{Al}_2\text{O}_3$ sample was smooth, which was caused by the adhesion of SiO_2 to Al_2O_3 (Fig. 1c and Supplementary Fig. 1b). Energy dispersive spectroscopy (EDS) provided evidence for the presence of SiO_x species (Supplementary Fig. 2), which was supported by the FT-IR band

near 1070 cm^{-1} corresponding to Si–O vibrations (Supplementary Fig. 3). Hexagonal plate crystals of ZSM-5 (approximately 2 μm in diameter) were randomly distributed on the Al_2O_3 support (Fig. 1d). The crystalline structure of ZSM-5 on the surface of the Al_2O_3 beads was further characterized by transmission electron microscopy (TEM). Analysis of the low-resolution images (Fig. 1e) showed a hexagonal plate-like morphology of the ZSM-5 component and that it was bonded with the amorphous Al_2O_3 component. The clear lattice fringes of these plate-like crystals (Fig. 1f) provided evidence of high crystallinity, and the interplanar spacing was 1.11 nm, which corresponds to the 101 lattice plane of ZSM-5²⁶. Furthermore, the selected-area electron diffraction (SAED) pattern revealed features characteristic of a plate-like single crystal (Fig. 1g)^{26,27}. The TEM-EDS mapping images (Fig. 1h) reveal an even distribution of Si, Al, and O in ZSM-5. The EDS analysis (Supplementary Fig. 4) was consistent with a silicon-to-aluminum molar ratio of approximately 100 for the ZSM-5 sample and was supported by the silicon-to-aluminum feed ratio used in the synthesis.

Characterization of catalysts

The X-ray diffraction (XRD) patterns of the zeolite-loaded aluminum-bead catalyst (Fig. 2a) show characteristic peaks at 8.0° , 8.9° , 23.3° , 24.0° , and 24.4° for the ZSM-5 zeolite (PDF#44-0003)²⁸ and at 45.9° and 66.8° for Al_2O_3

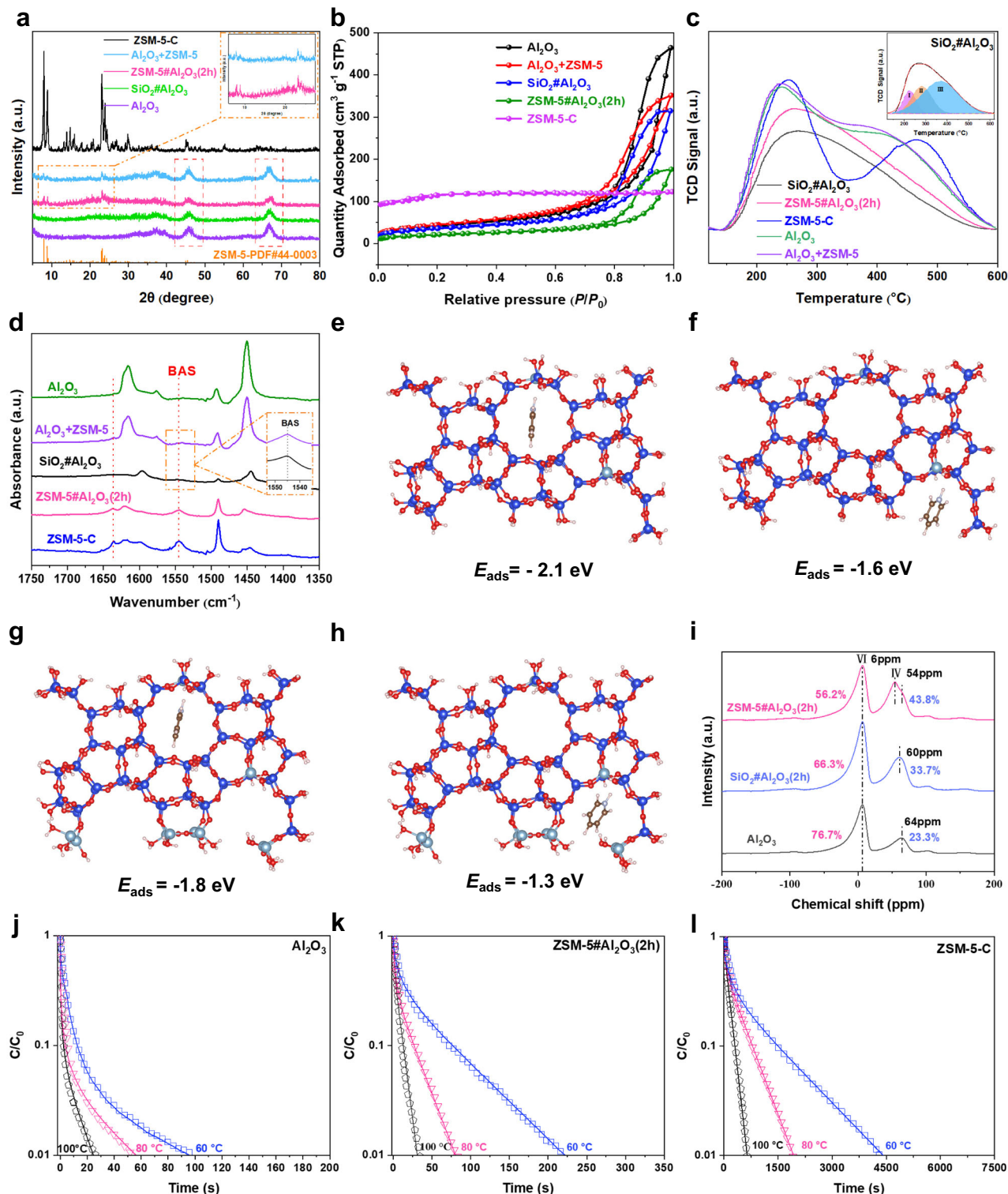


Fig. 2 | Characterization of the catalysts. a XRD patterns of ZSM-5-C, $\text{Al}_2\text{O}_3 + \text{ZSM-5}$, $\text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$, $\text{SiO}_2\#\text{Al}_2\text{O}_3$, and Al_2O_3 . **b** N_2 adsorption-desorption isotherms of Al_2O_3 , $\text{Al}_2\text{O}_3 + \text{ZSM-5}$, $\text{SiO}_2\#\text{Al}_2\text{O}_3$, $\text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$, and ZSM-5-C. **c** NH_3 -TPD profiles of various catalysts. **d** Py-IR spectra of various catalysts. **e** Pyridine adsorption on Brønsted acid sites of ZSM-5. **f** Pyridine adsorption on Brønsted acid sites at the boundary of ZSM-5. **g** Pyridine adsorption

on Brønsted acid sites located far from the $\gamma\text{-Al}_2\text{O}_3$ junction. **h** Pyridine adsorption on Brønsted acid sites located close to the $\gamma\text{-Al}_2\text{O}_3$ junction. Atom colors: Si (blue), O (red), H (white), Al (light cyan), and C/N of pyridine (brown/gray). **i** ^{27}Al MAS NMR spectra of $\text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$, $\text{SiO}_2\#\text{Al}_2\text{O}_3$ and Al_2O_3 . **j**, **k**, **l** Diffusion curves of n -hexane. **j** Al_2O_3 . **k** $\text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$. **l** ZSM-5-C.

(PDF#10-0425)²⁹. The supported sample (ZSM-5#Al₂O₃(2 h)) had no impurity phase, and the crystallinity of ZSM-5 in ZSM-5#Al₂O₃(2 h) was similar to that in the physical mixture (Al₂O₃+ZSM-5) containing 6 wt% zeolite. Figure 2b shows the nitrogen sorption isotherms of the samples. Al₂O₃ and ZSM-5-C showed typical type IV and type I isotherms, respectively, consistent with literature findings^{30,31}. All Al₂O₃-based materials contained a hysteresis loop at $0.6 < P/P_0 < 1$. Compared with Al₂O₃, the hysteresis loop of ZSM-5#Al₂O₃(2 h) exhibited a significant reduction, indicating that the growth of zeolite on Al₂O₃ occurred with both pore plugging and subsequent crystal growth. However, the isotherm of Al₂O₃ + ZSM-5 retained the structural characteristics of both ZSM-5 and Al₂O₃. The smaller hysteresis loop results from the reduced Al₂O₃ content. In the corresponding pore size distribution curves, one peak was observed with a mean size of 32 nm for ZSM-5#Al₂O₃(2 h) (Supplementary Fig. 5 and Supplementary Discussion 1). The detailed texture characteristics of the tested specimens are provided in Supplementary Table 1.

The acidic properties of all the materials were characterized by NH₃-TPD (Fig. 2c). The ZSM-5#Al₂O₃(2 h) and SiO₂#Al₂O₃ catalysts exhibited a single tailing peak at 260 °C, distinguishing them from the other three samples. Information on acidity and its distribution is presented in Supplementary Table 2. The ZSM-5 zeolite content in the supported sample was similar to that in the physically mixed sample, but the latter exhibited a higher total acid amount. This results from Al₂O₃ providing numerous acidic sites in physically mixed catalysts. Compared with the Al₂O₃ + ZSM-5 sample, the peak position of the strong acidic sites in the ZSM-5#Al₂O₃(2 h) sample shifted to lower temperatures, suggesting that the strong acidic sites of the supported catalyst were weaker than those of the physically mixed catalyst. The Py-IR spectra (Fig. 2d) provided information on the Brønsted acid sites (BAS) and Lewis acid sites (LAS)⁴. Except for the Al₂O₃ sample, all the catalysts had both BAS and LAS. At elevated outgassing temperatures, pyridine molecules tend to desorb from weak LAS, whereas they remain firmly coordinated to stronger BAS due to their higher binding energies. This results in a higher BAS/LAS (B/L) ratio³². The B/L ratio declined progressively with increasing outgassing temperature, suggesting that the BAS on the catalyst were predominantly weak- or moderate-strength. As summarized in Supplementary Table 3, the B/L acidity ratio for the ZSM-5-C sample increased with increasing outgassing temperature, whereas that of the ZSM-5#Al₂O₃(2 h) sample decreased. These results indicate that ZSM-5-C is characterized by a high concentration of strong BAS, whereas the acidic site distribution of the ZSM-5#Al₂O₃(2 h) sample is dominated by weak- and moderate-strength BAS. Combined with the data results of the SiO₂#Al₂O₃ catalyst, the weak and moderate BAS may be provided by silanol groups³³. Notably, the ZSM-5#Al₂O₃(2 h) catalyst exhibited a higher density of strong BAS than the Al₂O₃ + ZSM-5 catalyst with an equivalent zeolite content (Supplementary Table 3). The strong BAS in the ZSM-5#Al₂O₃(2 h) catalyst may be composed of three parts: ZSM-5 zeolite units, the interfacial effect between the ZSM-5 zeolite and Al₂O₃²², and the interfacial effect between SiO₂ and Al₂O₃. The SiO₂#Al₂O₃ catalyst contained a small amount of strong BAS, confirming the interface effect between Al₂O₃ and SiO₂ (Supplementary Table 3).

DFT calculations were conducted to investigate how the integration of γ -Al₂O₃ affects the Brønsted acidity strength of ZSM-5 by evaluating the adsorption energies of pyridine at identical T-sites located far from or adjacent to the γ -Al₂O₃ layer (Supplementary Fig. 6). Results show that the acid strength in the composite is reduced by approximately 0.3 eV relative to that of pristine ZSM-5 (Fig. 2e–h), demonstrating that the integration of γ -Al₂O₃ intrinsically weakens the Brønsted acidity strength of ZSM-5.

The ²⁷Al MAS NMR spectrum of ZSM-5#Al₂O₃(2 h) is depicted in Fig. 2i. The chemical shift at 6 ppm was observed for all samples and was assigned to octahedrally coordinated Al species (Al^{VI})³⁴. The chemical shift at about 54 ppm was assigned to tetrahedral framework aluminum (Al^{IV})³⁵. For ZSM-5#Al₂O₃(2 h), the chemical shift was 10 ppm (Al^{IV} band), consistent with framework aluminum in the zeolite on Al₂O₃. This was supported by the slight tendency of the double ²⁷Al NMR band for the Al^{IV} sites in the ZSM-5#Al₂O₃(2 h) catalyst. The relative fraction of Al^{VI}, as

determined from the integral intensities, decreased from 76.7% (Al₂O₃) to 66.3% (SiO₂#Al₂O₃), and to 56.2% (ZSM-5#Al₂O₃(2 h)). This trend was related to the presence of ZSM-5 and polycondensation between Al₂O₃ and SiO₂, leading to the formation of Al^{IV}³⁶.

The time-dependent desorption curves for *n*-hexane from zero-length column (ZLC) chromatography and the corresponding diffusion activation energy plots are shown in Fig. 2j–l. In the temperature-dependent family of time-dependent desorption curves, the data (symbols) were aligned very closely to the model used in the regression analysis (lines, Crank diffusion model³⁷). From this, the temperature-dependent apparent diffusion coefficients (D_{eff}/R^2) were derived and are presented in Supplementary Table 4. The diffusion activation energies of *n*-hexane followed the sequence Al₂O₃ < ZSM-5#Al₂O₃(2 h) < ZSM-5, with the unsupported ZSM-5 displaying significantly higher values (Supplementary Fig. 7). Hence, the contribution of mesopores to the diffusion of *n*-alkanes indicates that the increase in the activation energy is correlated with the decrease in mesopore volume (Supplementary Table 1)³⁸. In addition, the increase in the BAS of the catalyst is expected to inhibit the diffusion of *n*-hexane³⁹.

Catalytic performance

In conjunction with the literature⁴⁰, catalytic degradation of polyolefins ($T > 450$ °C) is accompanied by pyrolysis. Therefore, it is important to mitigate LDPE pyrolysis. From the LDPE conversion (Supplementary Fig. 8), it can be noted that LDPE pyrolysis is negligible between 200 °C and 300 °C. The optimized reaction conditions were determined by conducting additional experiments (Supplementary Fig. 9 and Supplementary Discussion 2). The impact of the initial N₂ pressure on cracking activity and resulting product distribution is detailed in Supplementary Discussion 3. A low-temperature melting catalysis method was used based on the melting characteristics of LDPE (~122 °C, as shown in Supplementary Fig. 11) and the strong capability of solid acid catalysts to break C–C bonds⁴. The conversion of the four samples followed the order: ZSM-5-C (92.4%) > ZSM-5#Al₂O₃(2 h) (89.9%) > Al₂O₃ + ZSM-5 (73.6%) > Al₂O₃ (1.8%) (Fig. 3a). With Al₂O₃, the conversion resembled that of the free-catalyst pyrolysis, and trace amounts of C₁–C₅ hydrocarbons were detected (Supplementary Fig. 12). This suggests that LAS alone possess insufficient catalytic activity to effectively promote the cracking of polyolefins (Supplementary Table 3)⁴¹. The ZSM-5-C catalyst exhibited the highest gas yield among the catalysts tested. The increased gas product yield results from the catalyst's pore structure and acidic properties acting in concert²³. The ZSM-5 zeolite constitutes the main active phase in all samples, with its intrinsic MFI framework topology and pore architecture remaining consistently preserved. In contrast to the other catalysts, ZSM-5-C is characterized by a high concentration of strong BAS (Supplementary Table 3). Consequently, the elevated gaseous product yield observed for ZSM-5-C is primarily ascribed to the intensified secondary cracking of hydrocarbon intermediates promoted by these strong acid centers. With this catalyst, the liquid product consisted predominantly of monocyclic aromatic hydrocarbons of C₇–C₉ (the yield was 42.4% (Fig. 3b and Supplementary Fig. 13a)). This is related to the specific pore structure and strong BAS within the pores of ZSM-5². The aromatics (C₇–C₉) in the liquid product were mainly composed of toluene, xylene, ethyltoluene, and trimethylbenzene (Supplementary Table 5 and Supplementary Fig. 14). Therefore, the importance of using catalysts with BAS is highlighted in this study. We correlated the LDPE conversion rate with the BAS density (Fig. 3c). The catalytic activity exhibited a positive correlation with the density of BAS. For the Al₂O₃ + ZSM-5, a reduction in conversion efficiency from 92.4% to 73.6% was observed compared with that of the ZSM-5-C catalyst. This was consistent with the decrease in the active component (ZSM-5 with strong BAS). The yields of olefins (11.9%) and alkanes (21.0%) in the liquid product slightly increased (Fig. 3b and Supplementary Fig. 13b), which may be related to the dispersion of ZSM-5 in the Al₂O₃ matrix²². Unlike the Al₂O₃ + ZSM-5 sample, the supported catalyst (ZSM-5#Al₂O₃(2 h)) promoted the formation of gasoline fractions (C₄–C₁₂ hydrocarbons) (Supplementary Fig. 13c). The liquid products were mainly olefins and alkanes, with yields of 27.9% and 35.8%, respectively,

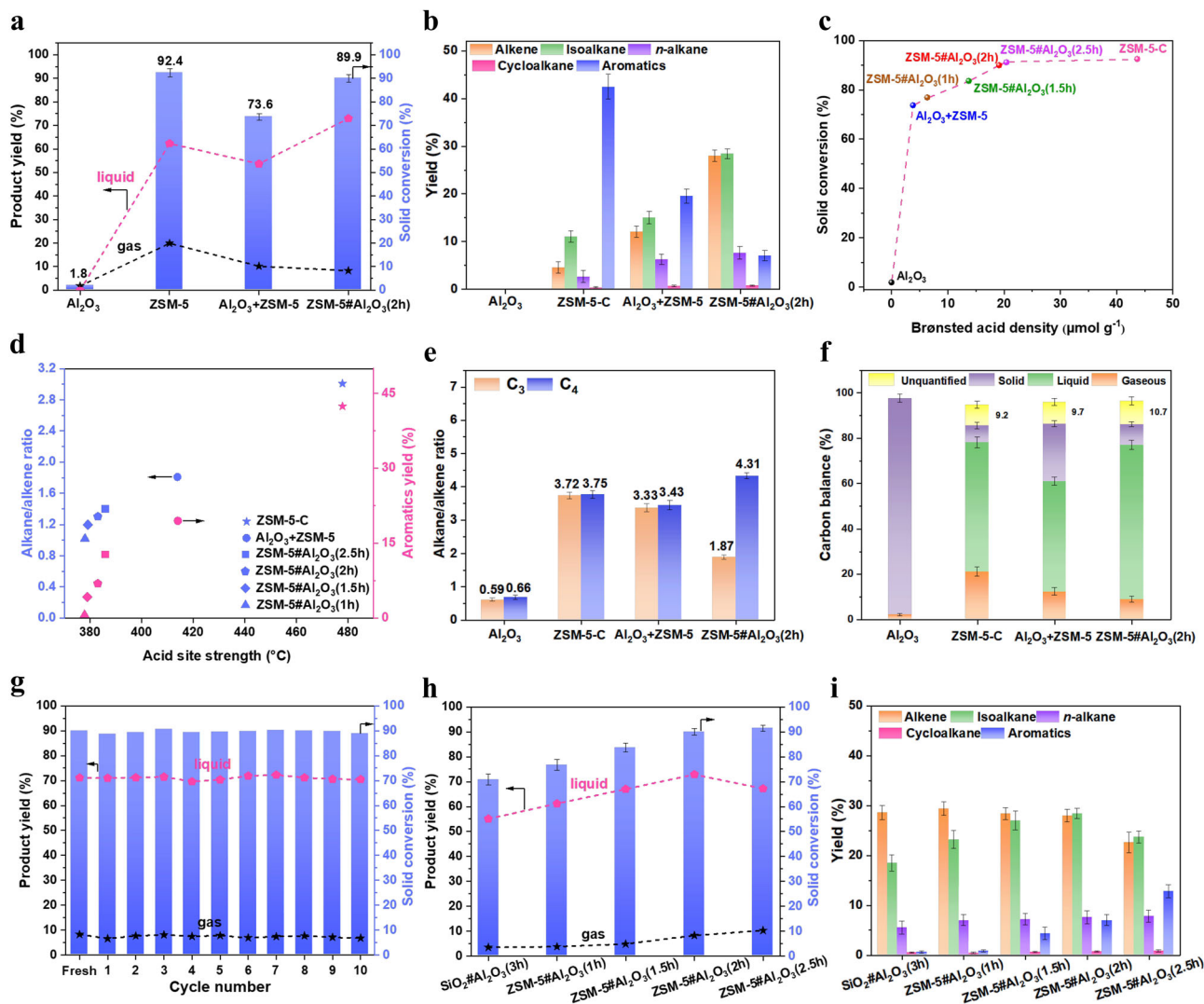


Fig. 3 | Catalytic performance of ZSM-5# Al_2O_3 (2 h) for LDPE upcycling. a LDPE conversion and product yields over various solid acid catalysts. Reaction conditions: LDPE powder, 0.5 g; catalyst, 0.5 g; temperature, 260 °C; N_2 , 0.1 MPa; reaction time: 1.5 h. **b** Product distribution in the liquid phase during LDPE upgrading with different catalysts. **c** Results of relating conversion to the Brønsted acid sites of the samples. **d** Results of correlating the alkane-to-olefin ratio and aromatic yield with the acid sites strength of the samples. Despite the inability of NH_3 -TPD to differentiate between BAS and LAS, the acid site desorption temperature increased upon zeolite addition. This shift suggests that these changes can be attributed to

modifications in the BAS. **e** The ratio of C_3 and C_4 alkanes/alkenes. **f** Carbon balance of the products across the four samples. **g** Product yield and LDPE conversion over ZSM-5# Al_2O_3 (2 h) catalyst in 10 cycles at 260 °C for 1.5 h. Between cycles, the sample was calcined in air at 550 °C. **h** LDPE conversion and product yields over samples with varying crystallization times. Reaction conditions: LDPE, 0.5 g; sample, 0.5 g; temperature, 260 °C; N_2 , 0.1 MPa; reaction time: 1.5 h. **i** Product distribution in the liquid phase during LDPE upgrading over samples with varying crystallization times.

while the yield of aromatics was only 6.9%. The enhanced catalytic efficiency stems from the additional BAS in the catalyst. These sites likely originated from the interfaces between Al_2O_3 , zeolite, and/or SiO_2 . Weak and moderate BAS were the most predominant (Supplementary Table 3). In addition, NH_3 -TPD and DFT calculations revealed that the acid strength of the ZSM-5# Al_2O_3 (2 h) sample was weaker relative to that of ZSM-5-C. Weak and moderate BAS preferentially catalyzed isomerization and type A β -scission processes, which are beneficial for the formation of branched hydrocarbons⁴². Moreover, the mesopores (32 nm) within the catalyst facilitated the diffusion of intermediates. The ZSM-5# Al_2O_3 (2 h) sample exhibited a lower diffusion activation energy of 32.6 kJ mol⁻¹, in contrast to 75.2 kJ mol⁻¹ for the pure ZSM-5 sample (Supplementary Fig. 7). This supports the conclusion that the formation of chain hydrocarbon is promoted. The product distribution is determined by the strength and accessibility of the BAS⁴³. To elucidate how BAS strength modulates product distribution, we correlated the acid strength (desorption temperature of the

strong acid peak center obtained by Gaussian fitting in NH_3 -TPD) with the alkane-to-olefin ratio and aromatic yield (Fig. 3d). The ratio of alkanes to olefins and the aromatic yield showed a positive correlation with acid strength. The intrinsic pore architecture of the ZSM-5 zeolite provides the necessary spatial confinement to facilitate the shape-selective cyclization and dehydrogenation of intermediates into aromatic compounds². The aromatization process occurs through dehydrogenation and hydrogen transfer steps, involving cyclic hydrocarbon intermediates^{2,44}. The intensified acid strength effectively catalyzed both the dehydrogenation and hydrogen transfer pathways^{45,46}, thereby enhancing the yield of aromatic compounds⁴⁷. As the acid strength intensified, a substantial fraction of olefinic intermediates underwent cyclization, resulting in a progressive increase in the alkane-to-olefin ratio within the liquid product. According to previous literature⁴⁸, the hydrogen transfer index (HTI), which denotes the ratio of alkanes to olefins in the gas products, serves as an indicator of the catalysts' hydrogen transfer capability. The distribution of the gaseous

products (Supplementary Fig. 12) shows that much higher yields were observed for the alkane products than for the olefins, except for the Al_2O_3 catalyst. The HTI for C_3 products followed the order of $\text{ZSM-5-C} > \text{Al}_2\text{O}_3 + \text{ZSM-5} > \text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$. This trend originates from the synergistic effects of the unique topological constraints of the MFI framework⁴⁹ and the distinct variations in acid site strength among the catalysts⁵⁰. These results demonstrate that, in ZSM-5-containing systems, an increase in acid strength promotes the hydrogen transfer reaction, thereby facilitating alkane formation^{45,46}. The significantly higher HTI of the C_4 product compared to the C_3 product over $\text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$ suggests a potential correlation with the strength of the BAS⁵¹ (Fig. 3e). The C_4 hydrocarbons underwent secondary cracking into C_3 species and methane at the strong Brønsted acid sites⁵², resulting in comparable HTI values for the C_3 and C_4 products over both the ZSM-5-C and $\text{Al}_2\text{O}_3 + \text{ZSM-5}$ samples. While the ZSM-5-C and $\text{Al}_2\text{O}_3 + \text{ZSM-5}$ samples featured strong BAS as their principal acidic component, the $\text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$ sample exhibited a prevalence of weak and moderate-strong BAS. The $\text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$ sample exhibited the largest fraction of unquantified components, owing to its high yield of volatile compounds ($\text{C}_4\text{--C}_6$) (Fig. 3f and Supplementary Fig. 13c). Furthermore, the $\text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$ catalyst was synthesized through direct crystallization without requiring molding, which made it easily recyclable and resistant to loss, with only a 1.6% decrease in mass after three cycling tests (Supplementary Table 6). The solid residue (unreacted or partially reacted LDPE, catalyst, and coke) obtained after the reaction was subjected to thermogravimetric analysis (TGA) (Supplementary Fig. 15 and Supplementary Discussion 4). The $\text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$ catalyst exhibited a lower coke content (5.6%) than the $\text{Al}_2\text{O}_3 + \text{ZSM-5}$ catalyst (8.8%). This result reveals that the textural properties of the catalyst are key to preventing coke formation. A comprehensive characterization of the carbonaceous deposits is provided in Supplementary Discussion 4. The reusability of $\text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$ was evaluated (Fig. 3g). The catalyst retained > 88% conversion and > 70% liquid yield after ten regeneration cycles, demonstrating high cracking activity and durability. XRD, nitrogen sorption analysis, NH_3 -TPD, and Py-IR confirmed the structural stability and acidic property stability of the catalyst after regeneration (Supplementary Figs. 19–21 and Supplementary Tables 7 and 8). The robust stability of the $\text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$ catalyst was further validated through a series of non-regeneration cycles, as detailed in Supplementary Discussion 5. The evolution process and formation mechanism of $\text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$ are detailed in Supplementary Discussion 6. The product yields and oil compositions across the catalysts ($\text{SiO}_2\#\text{Al}_2\text{O}_3(3\text{ h})$, and $\text{ZSM-5}\#\text{Al}_2\text{O}_3(1\text{ h}, 1.5\text{ h}, 2\text{ h}, \text{ and } 2.5\text{ h})$) are illustrated in Fig. 3h, i. The LDPE conversion rate and gaseous product yields gradually increased with an increase in crystallization time, which was positively correlated with the concentration of BAS (Supplementary Table 13). Interestingly, $\text{ZSM-5}\#\text{Al}_2\text{O}_3(2.5\text{ h})$ exhibited the highest BAS concentration of all catalysts tested (Supplementary Table 13), but its oil yield was only 67.2%, while the yield of aromatics was as high as 12.7%. This phenomenon is ascribed to the elevated concentrations of BAS and LAS (Supplementary Table 13), which facilitated aromatization and enhanced the secondary cracking of short-chain intermediates into gaseous products^{15,44,53}. Composite catalysts with prolonged zeolite crystallization time promoted a higher yield of alkanes in the liquid products, owing to enhanced BAS strength in the catalysts. The elevated desorption temperature observed for the catalysts provides further evidence for this conclusion (Supplementary Table 12). These results also indicate a synergy between strong acid sites and pore structure, which in turn leads to an increased aromatic yield for the $\text{ZSM-5}\#\text{Al}_2\text{O}_3(\text{Xh})$ catalysts with a longer preparation time. The HTI for the C_3 and C_4 fractions obtained over the $\text{ZSM-5}\#\text{Al}_2\text{O}_3(\text{Xh})$ series is illustrated in Supplementary Fig. 37. As the crystallization time was extended, the HTI of the C_3 products exhibited a monotonic increase, whereas the C_4 products displayed a distinct trend of initial increase followed by a subsequent decrease. This behavior is ascribed to the progressive increase in both the total and strong Brønsted acid site densities within the synthesized samples with prolonged crystallization time (Supplementary Table 13). It was also noted that the selectivity of gasoline-like

fuel ($\text{C}_4\text{--C}_{12}$) in the liquid product of the four $\text{ZSM-5}\#\text{Al}_2\text{O}_3(\text{Xh})$ catalysts was 97–99% (Supplementary Table 14). The prediction of the research octane number (RON) (Supplementary Table 15) showed that the $\text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$ catalyst exhibited the highest RON (88.3). The liquid product obtained using the $\text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$ catalyst exhibited a lower aromatic content than that of commercial gasoline (9.6% versus 32.5%) (Supplementary Table 15). To demonstrate the superior performance of the synthesized catalysts relative to zeolite Y, a comprehensive comparative analysis is provided in Supplementary Discussion 9. As summarized in Supplementary Table 17, the $\text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$ catalyst outperforms previously reported ZSM-5-based materials, achieving the highest yield of gasoline-range hydrocarbons under relatively milder conditions (<280 °C).

Practical applications of PE catalytic cracking and a study of the reaction mechanism

In this context, we also conducted catalytic tests using virgin granules of LDPE, HDPE, and LDPE + HDPE, as well as plastic wrap, bottles, and transparent ziplock bags (Fig. 4a) with the optimized catalyst ($\text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$). Commercial-grade PE products typically contain a complex matrix of polymeric constituents, additives, and stabilizers, as detailed in Supplementary Discussion 10. The melting points of these plastics were determined (Supplementary Fig. 44). The $\text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$ catalyst effectively converted all plastic waste into liquid products with high yields (60–70%) (Fig. 4b). Furthermore, the catalytic degradation of polyvinyl chloride (PVC) and polypropylene (PP) was investigated to evaluate the versatility of the catalyst, with results detailed in Supplementary Discussion 11.

The thermogravimetric curves of pure LDPE and the catalyst-LDPE mixture samples are shown in Fig. 4c. The pyrolysis of pure LDPE started at approximately 402 °C, indicating that the C–C bonds of LDPE were difficult to scission at a low temperature (260 °C) in this study. The introduction of the catalysts lowered the temperature of thermal degradation. LDPE decomposed at a lower temperature over the $\text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$ composite than over the $\text{Al}_2\text{O}_3 + \text{ZSM-5}$ physical mixture, demonstrating the composite's superior catalytic activity⁵⁴. These results validate the catalytic potential of ZSM-5 for low-temperature LDPE degradation. Two mass loss events of the $\text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$ induced degradation of LDPE were identified by fitting the DTG curve (Fig. 4d). The low temperature peak can be attributed to catalytic cracking (carbonium-ion mechanism), whereas the high temperature peak may be assigned to thermal cracking (free-radical mechanism)¹⁸.

The possible reaction pathway for the catalytic degradation of LDPE over $\text{ZSM-5}\#\text{Al}_2\text{O}_3(2\text{ h})$ is presented in Fig. 4e. In contrast to the free radical mechanism observed in thermal cracking (>400 °C), it can be noted that catalytic cracking predominantly operates through the carbenium-ion mechanism^{55,56}. To further discuss the mechanism of catalytic cracking, it can be noted that in the heating stage, LDPE first melted without reacting (at a temperature of ~122 °C) (Supplementary Fig. 11). At an increased temperature, the long-chain polyolefins seemed to have first adsorbed onto the BAS of the $\text{SiO}_2\#\text{Al}_2\text{O}_3$ part of composite sample and on the external surface of ZSM-5 part to form carbenium ion intermediates. This triggered a cascade of reactions (skeletal rearrangement and β -scission) that produced a large number of short-chain hydrocarbons. Type A β -scission exhibits a relatively low activation barrier and a high reaction rate compared to the other three types of β -scissions (types B₁, B₂, and C)⁴². It can be observed that the proportion of branched hydrocarbon (*i*-alkanes and *i*-alkenes) in the $\text{C}_4\text{--C}_{12}$ fraction exceeded 69% (Supplementary Table 18). This indicates that type A β -scission (involving tertiary carbenium ions) was dominant⁴². Carbenium ions undergo cyclization and dehydroaromatization (in the micropores of the ZSM-5 zeolite) to form aromatic compounds when positioned at a suitable distance (2–4 atoms) from adjacent C=C bonds^{2,57,58}. In addition, it has been established that branched chains of polyolefins can undergo end-chain scission, directly forming light hydrocarbon products¹⁵. The synergistic effects of carbocation chemistry and diffusion seem to have enabled the selective production of $\text{C}_4\text{--C}_{12}$ hydrocarbons under mild

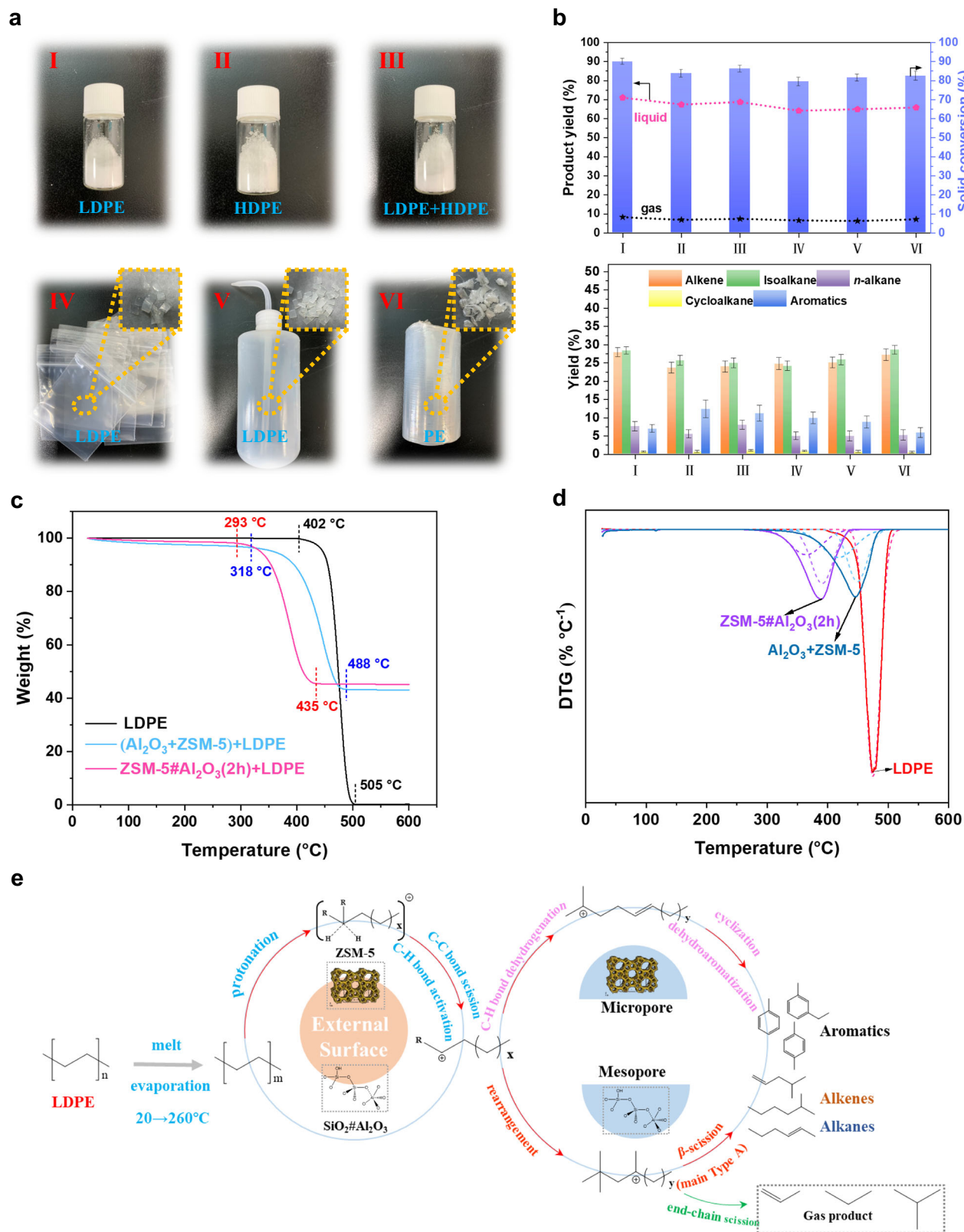


Fig. 4 | Practical applications of PE cracking and the proposed reaction mechanism. a Photograph of various types of PE plastics. **b** Conversion and product distribution for different PE plastics over the ZSM-5/Al₂O₃ (2 h) sample. Reaction conditions: LDPE, 0.5 g; catalyst, 0.5 g; temperature, 260 °C; N₂, 0.1 Mpa; reaction time: 1.5 h. TG (c) and DTG (d) curves of pure LDPE, LDPE + (Al₂O₃+ZSM-5) and

LDPE + (ZSM-5/Al₂O₃(2 h)) under inert N₂. The decomposition of the LDPE + (ZSM-5/Al₂O₃(2 h)) sample can be roughly divided into three stages: slight decomposition (<293 °C), rapid decomposition (293–435 °C), and termination of decomposition (435 °C). **e** Schematic of the reaction pathways for LDPE catalytic pyrolysis over ZSM-5/Al₂O₃(2 h).

conditions (260 °C) in the zeolite compartment of the catalyst. The process was performed in a 1 L CSTR (Continuous Stirred-Tank Reactor) unit.

Conclusion

A zeolite- Al_2O_3 -supported catalyst (ZSM-5# Al_2O_3 (2 h)) was prepared using a hydrothermal synthesis strategy that included a pretreatment step of the Al_2O_3 beads with SiO_x species to bond and control the loading of ZSM-5. The catalyst had the characteristics of no molding, easy recovery, and resistance to loss, and could directly upgrade LDPE to gasoline-range hydrocarbons at low temperatures without the need for H_2 . More importantly, the interfacial interaction between Al_2O_3 and zeolite (or SiO_2) leads to the formation of additional BAS, thereby enhancing the catalyst's capability for PE cracking. Notably, the yield of the C_4 – C_{12} products was 19% higher than that obtained using a conventional physically mixed catalyst. The synergistic effect of the BAS and mesopores was proven to be the key factor affecting product distribution. Our research provides insights for designing catalysts to chemically convert plastic waste into valuable products under mild conditions.

Methods

Reagents and materials

Sodium hydroxide (NaOH, 96 wt%, Shanghai Macklin Biochemical Technology Co., Ltd.), tetrapropyl ammonium hydroxide (TPAOH, 25 wt%, aqueous solution, Beijing InnoChem Science & Technology Co., Ltd.), silica sol (SiO_2 , 30 wt%, Shanghai Macklin Biochemical Technology Co., Ltd.), sodium aluminate (NaAlO_2 , Al_2O_3 , 41 wt%, Shanghai Aladdin Biochemical Technology Co., Ltd.), and alumina beads (γ - Al_2O_3 , particle size: ~2 mm, China Research Institute of Daily Chemistry Co., Ltd.). Low-density polyethylene powder (LDPE, with an average molecular weight of 50,000) and high-density polyethylene powder (HDPE, with an average molecular weight of 80,000) were purchased from Sinopec Group.

Preparation of catalysts

The Al_2O_3 beads were immersed in a 1 mol L^{-1} NaOH aqueous solution at room temperature and subjected to ultrasonication for 20 min. Subsequently, the Al_2O_3 beads were washed several times until neutral. The drying process was conducted at 100 °C for 8 h. The substrate was denoted as Na- Al_2O_3 .

The ZSM-5# Al_2O_3 catalysts were prepared using a two-step in situ hydrothermal synthesis method. The ZSM-5 zeolite precursor solution was obtained by aging a gel with a molar ratio of 1 Al_2O_3 :200 SiO_2 :30 TPA^+ :6890 H_2O at 25 °C for 12 h. The aqueous solution of silica was similar to the zeolite precursor solution, except for the absence of the aluminum source (NaAlO_2) and template (TPAOH). In the initial hydrothermal synthesis, Na- Al_2O_3 beads (3.0 g) were mixed with an aqueous silica solution (36.0 g). The mixture was subsequently placed in a polytetrafluoroethylene container and treated at 180 °C for 3 h. Following cooling, the beads (denoted as SiO_2 # Al_2O_3) were washed until neutral. After complete drying, the SiO_2 # Al_2O_3 beads were subjected to thermal calcination at 550 °C for 6 h. In the second hydrothermal synthesis, SiO_2 # Al_2O_3 beads (1.8 g) were added to the zeolite precursor solution (36.0 g) and subjected to stirring for 5 min. The mixture was placed in a polytetrafluoroethylene container and subjected to a hydrothermal reaction at 180 °C for 2 h. The final product was labeled ZSM-5# Al_2O_3 (2 h).

For comparison, the zeolite precursor was crystallized without the addition of Na- Al_2O_3 beads using identical synthesis procedures employed for fabricating ZSM-5# Al_2O_3 . The sample was labeled as ZSM-5-C, where 'C' represents the control group. The effect of alumina bead pretreatment on crystallization was investigated, and the sample preparation procedures are described in the Supplementary Methods. To further elucidate the crystallization mechanism, intermediate products were obtained at specific time intervals during the synthesis process and designated as SiO_2 # Al_2O_3 (t) or ZSM-5# Al_2O_3 (t), where 't' denotes the corresponding synthesis duration. All samples were subjected to a heating ramp at a rate of 1 °C min^{-1} to 550 °C, followed by a 6-hour isothermal hold under an air atmosphere.

Finally, the H-type samples, labeled H-ZSM-5# Al_2O_3 (t) and H-ZSM-5-C, were obtained via ammonium ion exchange in a 1 mol L^{-1} NH_4Cl aqueous solution at 80 °C for 3 h (performed in triplicate), and subsequent calcination at 550 °C for 6 h.

In addition, a physically mixed catalyst (Al_2O_3 +ZSM-5) was obtained by mixing alumina with H-ZSM-5-C at a mass ratio of 16:1 (g g^{-1}). Here, the mass ratio of H-ZSM-5-C to alumina in Al_2O_3 + ZSM-5 was determined by the zeolite content close to the ZSM-5# Al_2O_3 (2 h) catalysts.

Catalytic test

The catalytic pyrolysis of LDPE over all catalysts was carried out in a 100 mL autoclave supplied by Parr Instrument Company. The LDPE powder and catalyst were mixed, placed into the reactor, and the reactor was then sealed hermetically. The Parr reactor was purged ten times with N_2 , then pressurized with N_2 to 0.1 MPa (atmospheric pressure) for the reaction. The temperature was ramped from 20 °C to 160 °C (half-power heating) and then rapidly increased to the target reaction temperature (full-power heating). All reactions were performed under an inert atmosphere. Upon completion of the reaction, the product was collected after cooling the Parr vessel to 10 °C.

Product analysis

Gaseous products were collected in aluminum foil sampling bags and analyzed via gas chromatography coupled with a flame ionization detector (GC-FID, Agilent 7820A) equipped with an Agilent HP-Plot GC column. The gas chromatogram was calibrated using the configured standard gas (CH_4 : 0.970%, C_2H_4 : 0.632%, C_2H_6 : 0.271%, C_3H_6 : 2.99%, C_3H_8 : 0.371%, n - C_4H_{10} : 0.749%, iso - C_4H_{10} : 0.795%, n - C_4H_{10} : 0.145%, iso - C_4H_{10} : 0.470%, n - C_5H_{12} : 0.070%, iso - C_5H_{12} : 0.244%, n - C_5H_{12} : 0.494%), and the absolute content of the gaseous product was determined by applying the relative mass correction factor.

The liquid oil was mixed with about 2 mL of dichloromethane (DCM, CH_2Cl_2) and analyzed by a GC-MS system using a Shimadzu QP2010SE Network GC fitted with a Restek Rxi-5Sil MS column. The standard curve method was used to quantify the hydrocarbon substances in the liquid²³.

The residues remaining after the reaction were collected with approximately 20–40 mL of n -hexane and transferred to a pre-weighed vial. The mixture of n -hexane and the residues was completely evaporated to dryness and then weighed.

The solid conversion of LDPE and the product yield (Y) were determined using Eqs. 1 and 2, respectively,

$$\text{Solid conversion} = [m(\text{LDPE}) + m(\text{catalyst}) - m(\text{residues})] / m(\text{LDPE}) \times 100\% \quad (1)$$

$$Y_i = m_i / m(\text{LDPE}) \times 100\% \quad (2)$$

where $m(\text{LDPE})$, $m(\text{catalyst})$, and $m(\text{residues})$ represent the masses of LDPE, the catalyst, and the solid residues, respectively, and m_i represents the mass of product i . The residues after the reaction consisted of the catalyst, unreacted polyethylene, coke, and solid products (wax). Therefore, the calculated conversion rate was underestimated, but the yield determination remained accurate.

Data availability

The authors declare that all data supporting the findings of this study are available within the paper and its Supplementary Information files.

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References

1. Rasmussen, S. C. From parkesine to celluloid: the birth of organic plastics. *Angew. Chem. Int. Ed.* **60**, 8012–8016 (2021).

2. Zhang, Z. et al. Highly selective upgrading of polyethylene into light aromatics via a low-temperature melting-catalysis strategy. *ACS Catal.* **14**, 2552–2561 (2024).
3. Deng, L. et al. Novel catalysts in catalytic upcycling of common polymer wastes. *Chem. Eng. J.* **471**, 144350 (2023).
4. Eschenbacher, A. et al. Highly selective conversion of mixed polyolefins to valuable base chemicals using phosphorus-modified and steam-treated mesoporous HZSM-5 zeolite with minimal carbon footprint. *Appl. Catal. B Environ.* **309**, 121251 (2022).
5. Dai, L. et al. Insights into structure–performance relationship in the catalytic cracking of high density polyethylene. *Appl. Catal. B Environ.* **318**, 121835 (2022).
6. Duan, D. et al. Synthesis of nanocrystalline cellulose induced hierarchical porous ZSM-5 for catalytic conversion of low-density polyethylene. *Fuel* **331**, 125757 (2023).
7. Xia, C. & Han, B. Plastics upcycling made more sustainable and practical. *Nat. Sustain.* **6**, 1518–1519 (2023).
8. Liu, J. et al. Catalytic pyrolysis of FeAlO_x and medical plastic waste: kinetic, slag conversion, and gas emission patterns. *J. Environ. Chem. Eng.* **12**, 112605 (2024).
9. Yang, G. et al. The catalytic pyrolysis of waste polyolefins by zeolite-based catalysts: a critical review on the structure–acidity synergies of catalysts. *Polym. Degrad. Stab.* **222**, 110712 (2024).
10. Chen, S. & Hu, Y. H. Chemical recycling of plastic wastes with alkaline earth metal oxides: a review. *Sci. Total Environ.* **905**, 167251 (2023).
11. Dai, L. et al. Pyrolysis technology for plastic waste recycling: a state-of-the-art review. *Prog. Energy Combust. Sci.* **93**, 101021 (2022).
12. Zhang, Z. et al. Recovering waste plastics using shape-selective nano-scale reactors as catalysts. *Nat. Sustain.* **2**, 39–42 (2019).
13. Dai, L. et al. Catalytic reforming of polyethylene pyrolysis vapors to naphtha range hydrocarbons with low aromatic content over a high silica ZSM-5 zeolite. *Sci. Total Environ.* **847**, 157658 (2022).
14. Qie, Z. et al. Catalytic pyrolysis of high-density polyethylene (HDPE) over hierarchical ZSM-5 zeolites produced by microwave-assisted chelation-alkaline treatment. *Fuel* **368**, 131532 (2024).
15. Hittinger, J. P. & Shantz, D. F. Systematic study of low temperature cracking of low-density polyethylene with ZSM-5. *Microporous Mesoporous Mater.* **343**, 112170 (2022).
16. Jumah, A. B., Tedstone, A. A. & Garforth, A. A. Hydrocracking of virgin and post-consumer polymers. *Microporous Mesoporous Mater.* **315**, 110912 (2021).
17. Wang, Z. et al. Tailor-made nanoseeds for the synthesis of zeolites with nanosized dimensions. *Cryst. Growth Des.* **23**, 6450–6460 (2023).
18. Lima, R. B. et al. Obtainment of hierarchical ZSM-5 zeolites by alkaline treatment for the polyethylene catalytic cracking. *Adv. Powder Technol.* **32**, 515–523 (2021).
19. Jia, X., Khan, W., Wu, Z., Choi, J. & Yip, A. C. K. Modern synthesis strategies for hierarchical zeolites: bottom-up versus top-down strategies. *Adv. Powder Technol.* **30**, 467–484 (2019).
20. Rocha, J. D. G. D. et al. Advances and environmental aspects on the synthesis of hierarchical zeolites revisited: a state-of-the-art description. *J. Environ. Chem. Eng.* **11**, 109397 (2023).
21. Peng, Y. et al. Conversion of low-density polyethylene into monocyclic aromatic hydrocarbons through continuous microwave pyrolysis with ex-situ dual-catalyst beds. *J. Clean. Prod.* **418**, 138039 (2023).
22. Wang, F. et al. Zeolite-modified alumina-bead catalyst for hierarchical cracking of bulky molecules. *Colloids Surf. A Physicochem. Eng. Asp.* **703**, 135316 (2024).
23. Cen, Z. et al. Upcycling of polyethylene to gasoline through a self-supplied hydrogen strategy in a layered self-pillared zeolite. *Nat. Chem.* **16**, 871–880 (2024).
24. Zhang, W. et al. Advances and challenges in low-temperature upcycling of waste polyolefins via tandem catalysis. *Angew. Chem. Int. Ed.* **64**, e202500559 (2025).
25. Ellis, L. D. et al. Chemical and biological catalysis for plastics recycling and upcycling. *Nat. Catal.* **4**, 539–556 (2021).
26. Zhang, J. et al. An efficient one-pot strategy for synthesizing hierarchical aluminosilicate zeolites using single structure directing agent. *Chem. Eng. J.* **335**, 822–830 (2018).
27. Bu, L. et al. Green synthesis of shaped nano-ZSM-5 toward more efficient methanol-to-hydrocarbon conversion. *Appl. Catal. A Gen.* **665**, 119393 (2023).
28. Liu, K. et al. Interplay between particle size and hierarchy of zeolite ZSM-5 during the CO₂-to-aromatics process. *ChemSusChem* **16**, e202300608 (2023).
29. Feng, R., Hu, X., Yan, X., Yan, Z. & Rood, M. J. A high surface area mesoporous γ -Al₂O₃ with tailoring texture by glucose template for ethanol dehydration to ethylene. *Microporous Mesoporous Mater.* **241**, 89–97 (2017).
30. Sun, J. et al. Rational design of poisoning-resistant catalyst based on γ -Al₂O₃ for hydrolysis of carbonyl sulfide. *Chem. Eng. Sci.* **295**, 120150 (2024).
31. Cui, X. et al. Selective aromatization of 1-hexene to BTX over core-shell structured Silicalite-1@ZSM-5 catalyst. *Sep. Purif. Technol.* **349**, 127881 (2024).
32. Jia, Y. et al. Nanocrystallite self-assembled hierarchical ZSM-5 zeolite microsphere for methanol to aromatics. *Microporous Mesoporous Mater.* **247**, 103–115 (2017).
33. Ardagh, M. A., Bo, Z., Nauert, S. L. & Notestein, J. M. Depositing SiO₂ on Al₂O₃: a route to tunable brønsted acid catalysts. *ACS Catal.* **6**, 6156–6164 (2016).
34. Zhao, Z. et al. Nature of five-coordinated Al in γ -Al₂O₃ revealed by ultra-high-field solid-state NMR. *ACS Cent. Sci.* **8**, 795–803 (2022).
35. Yuan, Z., Bai, Y., Gong, K. & Huang, W. Accurate measurements of NH₃ differential adsorption heat unveil structural sensitivity of Brønsted acid and Brønsted/Lewis acid synergy in zeolites. *J. Phys. Chem. Lett.* **15**, 863–868 (2024).
36. Xie, M. et al. Enhanced catalytic performance of the FCC catalyst with an alumina matrix modified by the zeolite Y structure-directing agent. *Ind. Eng. Chem. Res.* **58**, 5455–5463 (2019).
37. Eic, M. & Ruthven, D. M. A new experimental technique for measurement of intracrystalline diffusivity. *Zeolites* **8**, 40–45 (1988).
38. Liu, S., Luo, C., Deng, X. & Fang, Y. Toward rational design of narrowly-distributed mesopore on ZSM-22 zeolite for enhanced Pt dispersion and n-alkane isomerization performance. *Fuel* **328**, 125282 (2022).
39. Xu, D., Ma, J., Zhao, H., Liu, Z. & Li, R. Adsorption and diffusion of n-heptane and toluene over mesostructured ZSM-5 zeolitic materials with acidic sites. *Fluid Phase Equilib.* **423**, 8–16 (2016).
40. Dai, L. et al. Catalytic microwave-assisted pyrolysis of plastic waste to produce naphtha for a circular economy. *Resour. Conserv. Recycl.* **198**, 107154 (2023).
41. Pennel, M. L., Maurya, A. K., Ebrahim, A. M., Tassone, C. J. & Cargnello, M. Activity of silica-alumina for the conversion of polyethylene into tunable aromatics below pyrolytic temperatures. *ACS Sustain. Chem. Eng.* **11**, 12623–12630 (2023).
42. Han, W. et al. Production of branched alkanes by upcycling of waste polyethylene over controlled acid sites of SO₄/ZrO₂-Al₂O₃ catalyst. *Angew. Chem. Int. Ed.* **64**, e202417923 (2025).
43. Zhang, H., Ma, Y., Song, K., Zhang, Y. & Tang, Y. Nano-crystallite oriented self-assembled ZSM-5 zeolite and its LDPE cracking properties: effects of accessibility and strength of acid sites. *J. Catal.* **302**, 115–125 (2013).
44. Müller, S. et al. Hydrogen transfer pathways during zeolite catalyzed methanol conversion to hydrocarbons. *J. Am. Chem. Soc.* **138**, 15994–16003 (2016).

45. Auepattana-aumrung, C., Suriye, K., Jongsomjit, B., Panpranot, J. & Praserttham, P. Inhibition effect of Na⁺ form in ZSM-5 zeolite on hydrogen transfer reaction via 1-butene cracking. *Catal. Today* **358**, 237–245 (2020).
46. Zhao, Q., Li, F., Wang, Y., Liu, Y. & He, M. Acid strength controlled reaction pathways of propylene conversion under olefin cracking conditions over ZSM-5. *Fuel* **371**, 132077 (2024).
47. Ma, H. et al. Theoretical study on the influence of ZSM-5 zeolite with different structures for methanol to aromatics. *Microporous Mesoporous Mater.* **294**, 109838 (2020).
48. Corma, A., González-Alfaro, V. & Orchillés, A. V. The role of pore topology on the behaviour of FCC zeolite additives. *Appl. Catal. A Gen.* **187**, 245–254 (1999).
49. Al-Shafei, E. N. et al. Effect of zeolite structure and addition of steam on naphtha catalytic cracking to improve olefin production. *Fuel* **321**, 124089 (2022).
50. Abdalla, A., Arudra, P. & Al-Khattaf, S. S. Catalytic cracking of 1-butene to propylene using modified H-ZSM-5 catalyst: a comparative study of surface modification and core-shell synthesis. *Appl. Catal. A Gen.* **533**, 109–120 (2017).
51. Liu, Y. et al. Selective and controllable cracking of polyethylene waste by Beta zeolites with different mesoporosity and crystallinity. *Adv. Sci.* **11**, 2404426 (2024).
52. Shou, H. et al. Cracking of butane on a Pt/H-ZSM-5 catalyst in the presence of hydrogen. *Ind. Eng. Chem. Res.* **61**, 4763–4773 (2022).
53. Rane, N., Kersbulck, M., van Santen, R. A. & Hensen, E. J. M. Cracking of n-heptane over Brønsted acid sites and Lewis acid Ga sites in ZSM-5 zeolite. *Microporous Mesoporous Mater.* **110**, 279–291 (2008).
54. Kokuryo, S. et al. Defect engineering to boost catalytic activity of Beta zeolite on low-density polyethylene cracking. *Mater. Today Sustain.* **17**, 100098 (2022).
55. Dong, Z. et al. Understanding the structure–activity relationships in catalytic conversion of polyolefin plastics by zeolite-based catalysts: a critical review. *ACS Catal.* **12**, 14882–14901 (2022).
56. Liu, L. et al. Cascaded catalytic copyrolysis of microalgae and LDPE with Ni/Biochar and HZSM-5 for improving high-grade bio-oil production. *ACS Sustain. Chem. Eng.* **11**, 15964–15974 (2023).
57. Wang, W. et al. Catalytic conversion of polyethylene into aromatics with Pt/ZSM-5: insights into reaction pathways and rate-controlling step regulation. *J. Mater. Chem. A* **11**, 14933–14940 (2023).
58. Dorado, C., Mullen, C. A. & Boateng, A. A. Origin of carbon in aromatic and olefin products derived from HZSM-5 catalyzed co-pyrolysis of cellulose and plastics via isotopic labeling. *Appl. Catal. B: Environ.* **162**, 338–345 (2015).

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F.W.: Methodology, Investigation, Writing-original draft. Q.D.: DFT investigation. Y.C.L.: Investigation, Data curation. Z.X.L.: Investigation. W.M.H.: Data curation, Investigation. J.H.M.: Methodology, Investigation, Funding acquisition. H.J.J.: Methodology, DFT investigation. J.J.Z.: Investigation. N.H.: Methodology, Writing-review, Editing. R.F.L.: Conceptualization, Supervision, Review & editing, Project administration, Funding acquisition.

Competing interests

The authors declare no competing interests.

Additional information

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